

Leonardo De Andrade Santos

FPGA / Embedded Systems Engineer | Low-Latency DSP • VHDL • C/C++ • Python • Linux

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Professional Summary

FPGA, firmware and embedded systems engineer with 4+ years of experience across real-time systems, low-level C/C++ development, Linux-based engineering workflows and FPGA-oriented signal processing. Strong background in Digital Predistortion for RF systems, including Python algorithm prototyping, memory polynomial modelling and VHDL/fixed-point implementation. Comfortable translating mathematical models into hardware-aware, timing-conscious architectures and debugging performance-constrained systems where latency, reliability and correctness matter.

Technical Skills

FPGA & RTL: VHDL, FPGA design, RTL simulation, fixed-point arithmetic, pipelined datapaths, timing-aware implementation, resource-constrained design, FPGA verification

DSP & RF: Digital Signal Processing, Digital Predistortion, memory polynomial models, RF power amplifier behavioural modelling, nonlinear systems, real-time DSP, SDR concepts

Programming & Systems: C/C++, Python, Linux, shell scripting, embedded Linux, bare-metal concepts, communication interfaces, hardware/software integration, TCP/IP and UDP concepts

Tools: Git, CI/CD-oriented workflows, debugging, Python/NumPy, testbenches, model-to-RTL validation, Xilinx ISE, Vivado academic exposure, Cadence Genus/ASIC flow exposure

Professional Experience

Embedded & Software Engineer — IoTag Technology

Oct 2022 – Feb 2026

- Developed low-level C/C++ software for real-time embedded and Linux-based systems used in telemetry and equipment monitoring.
- Implemented communication interfaces, data acquisition pipelines and backend integrations for connected devices operating under field constraints.
- Debugged hardware/software interactions, communication failures and performance issues across deployed embedded systems.
- Built Python-based tooling for data analysis, validation and reliability checks over equipment datasets.

Engineering Intern — Lactec

2022

- Developed embedded systems and sensor-based solutions for measurement, validation, integration and hardware testing.

FPGA, DSP & Research Projects

Real-Time DSP and FPGA Implementation for RF Power Amplifier Linearization

- Designed and evaluated Digital Predistortion models for RF amplifier linearization using memory polynomial structures.
- Prototyped DSP algorithms in Python and translated them into fixed-point VHDL architectures for FPGA-oriented implementation.
- Built validation flows comparing Python fixed-point models with VHDL simulation outputs, focusing on bit-accurate behaviour and NMSE performance.
- Worked with complex arithmetic, delay lines, polynomial terms, coefficient quantization, truncation and pipelined datapath organisation.
- Developed FPGA-oriented architectures targeting low-latency, deterministic and real-time DSP processing.

FPGA-based TRNG / ASIC Research

- Worked on a true random number generator described in VHDL, including RTL understanding, modification and validation; used Cadence tools in a microelectronics research context.

DSP and Wireless Systems Research — GICS / UFPR

- Conducted research in signal processing for wireless systems, RF optimisation and hardware-aware DSP implementation; presented research at Seminários de Microeletrônica do Paraná.

Baja SAE — Electrical System

- Designed wiring harnesses and PCB-related subsystems; developed firmware and supported integration testing between electrical, mechanical and embedded subsystems.

Education

Master of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Expected Aug 2026

Research focus: RF power amplifier behavioural modelling, Digital Predistortion, memory polynomial models, real-time DSP systems and FPGA/VHDL implementation.

Bachelor of Electronic Engineering — Universidade Federal do Paraná (UFPR), Brazil

Relevant experience: digital systems, microelectronics, embedded systems, signal processing, firmware, PCB design and electrical systems.